

Plastic Deformation Energy UMAT

Heat caused by **plastic energy dissipation** is calculated from the **plastic work** done during permanent deformation. When a material yields, part of the mechanical work converts into heat.

Core Formula

The heat generated per unit volume is usually:

$$q = \beta \int \sigma d\varepsilon_p$$

Where:

- q = heat generated per unit volume (J/m³)
 - β = Taylor–Quinney coefficient (fraction converted to heat, usually **0.85 to 0.95** for metals)
 - σ = stress during plastic deformation (Pa)
 - $d\varepsilon_p$ = incremental plastic strain
-

Simple Approximation (constant flow stress)

If stress is roughly constant during yielding:

$$q = \beta \sigma \varepsilon_p$$

Where:

- σ = average plastic flow stress
 - ε_p = accumulated plastic strain
-

Temperature Rise

If you want resulting temperature increase:

$$\Delta T = \frac{q}{\rho c_p}$$

Where:

- ρ = density (kg/m³)
- c_p = specific heat (J/kg·K)

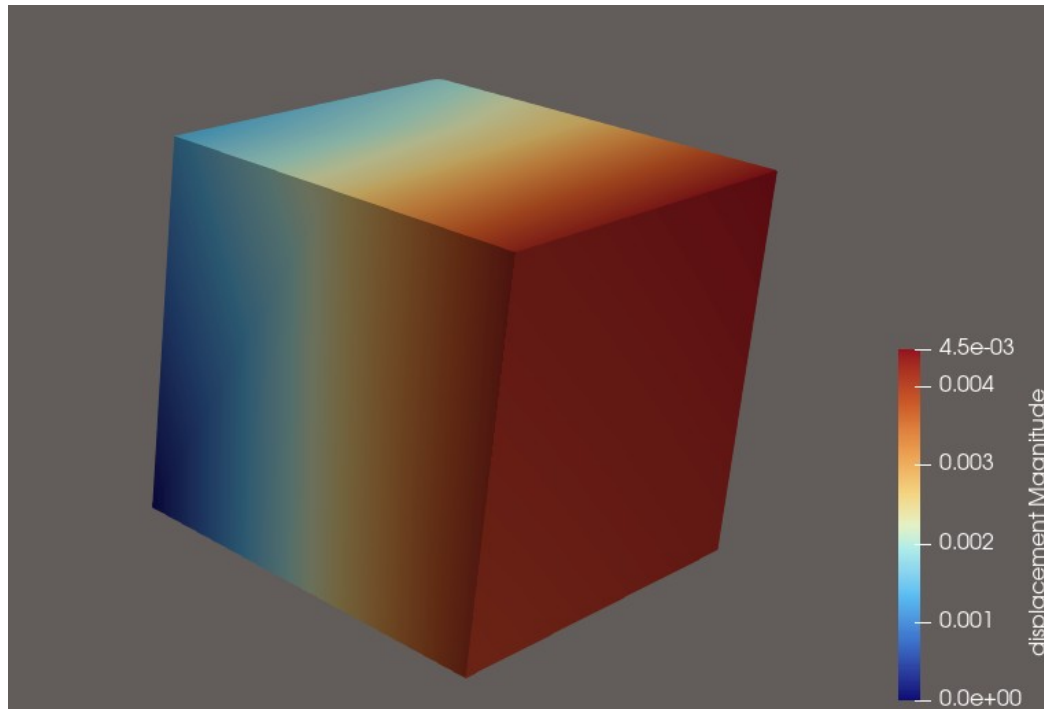
Example Problem

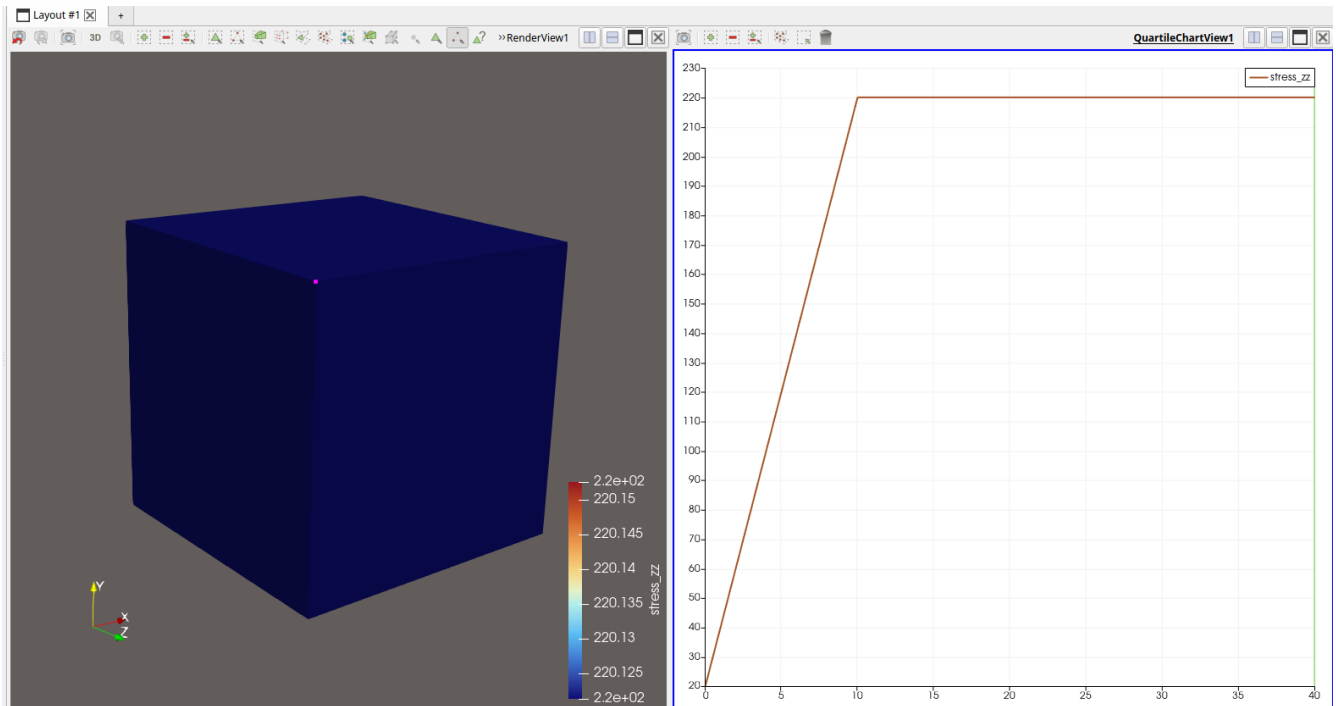
A 1 x 1 x 1 steel block is put into uniaxial tension by restraining one end and displacing the other end as a function of time. $E = 200000.0$, $\nu = 0.30$, the plastic model is perfectly plastic at a yield stress of 220.0. The material density is 7850. and the specific heat capacity $C_p = 470$.

A spreadsheet was developed to calculate the close formed solution to provide a test case.

Step	Total Strain	Stress (MPa)	Elastic Strain	Plastic Strain	Plastic Work Inc (MPa)	Cum Work (MPa)	Heat Inc (J/m3)	Cum Heat (J/m3)	Temp Rise Inc (C)	Cum Temp Rise (C)
1	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
2	0.00010	20.00100	0.00010	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
3	0.00020	40.00400	0.00020	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
4	0.00030	60.00900	0.00030	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
5	0.00040	80.01600	0.00040	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
6	0.00050	100.02500	0.00050	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
7	0.00060	120.03600	0.00060	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
8	0.00070	140.04900	0.00070	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
9	0.00080	160.06400	0.00080	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
10	0.00090	180.08100	0.00090	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
11	0.00100	200.10000	0.00100	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
12	0.00110	220.12100	0.00110	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
13	0.00120	220.12101	0.00110	0.00010	0.02204	0.02204	19833.66393	19833.66393	0.00538	0.00538
14	0.00130	220.12102	0.00110	0.00020	0.02204	0.04408	19835.64641	39669.31034	0.00538	0.01075
15	0.00140	220.12103	0.00110	0.00030	0.02204	0.06612	19837.62363	59506.93397	0.00538	0.01613
16	0.00150	220.12104	0.00110	0.00040	0.02204	0.08816	19839.61176	79346.54573	0.00538	0.02151
17	0.00160	220.12105	0.00110	0.00050	0.02205	0.11021	19841.59035	99188.13607	0.00538	0.02688
18	0.00170	220.12106	0.00110	0.00060	0.02205	0.13226	19843.57233	119031.70840	0.00538	0.03226
19	0.00180	220.12107	0.00110	0.00070	0.02205	0.15431	19845.55500	138877.26340	0.00538	0.03764
20	0.00190	220.12108	0.00110	0.00080	0.02205	0.17636	19847.53768	158724.80109	0.00538	0.04302
21	0.00200	220.12109	0.00110	0.00090	0.02206	0.19842	19849.51968	178574.32076	0.00538	0.04840
22	0.00210	220.12110	0.00110	0.00100	0.02206	0.22047	19851.50029	198425.82105	0.00538	0.05378
23	0.00220	220.12111	0.00110	0.00110	0.02206	0.24253	19853.48160	218279.30265	0.00538	0.05916
24	0.00230	220.12112	0.00110	0.00120	0.02206	0.26459	19855.46634	238134.76899	0.00538	0.06454
25	0.00240	220.12113	0.00110	0.00130	0.02206	0.28666	19857.44627	257992.21526	0.00538	0.06993
26	0.00250	220.12114	0.00110	0.00140	0.02207	0.30872	19859.42964	277851.64491	0.00538	0.07531
27	0.00260	220.12115	0.00110	0.00150	0.02207	0.33079	19861.41164	297713.05654	0.00538	0.08069
28	0.00270	220.12116	0.00110	0.00160	0.02207	0.35286	19863.39156	317576.44810	0.00538	0.08608
29	0.00280	220.12117	0.00110	0.00170	0.02207	0.37494	19865.37563	337441.82373	0.00538	0.09146
30	0.00290	220.12118	0.00110	0.00180	0.02207	0.39701	19867.35624	357309.17997	0.00538	0.09684
31	0.00300	220.12119	0.00110	0.00190	0.02208	0.41909	19869.34031	377178.52027	0.00539	0.10223
32	0.00310	220.12120	0.00110	0.00200	0.02208	0.44117	19871.31884	397049.83911	0.00539	0.10762
33	0.00321	220.12121	0.00110	0.00210	0.02208	0.46325	19873.30291	416923.14203	0.00539	0.11300
34	0.00331	220.12122	0.00110	0.00220	0.02208	0.48533	19875.28491	436798.42694	0.00539	0.11839
35	0.00341	220.12123	0.00110	0.00231	0.02209	0.50742	19877.26552	456675.69245	0.00539	0.12378
36	0.00351	220.12124	0.00110	0.00241	0.02209	0.52951	19879.25168	476554.94414	0.00539	0.12917
37	0.00361	220.12125	0.00110	0.00251	0.02209	0.55160	19881.23021	496436.17434	0.00539	0.13455
38	0.00371	220.12126	0.00110	0.00261	0.02209	0.57369	19883.21568	516319.39003	0.00539	0.13994
39	0.00381	220.12127	0.00110	0.00271	0.02209	0.59578	19885.19350	536204.58353	0.00539	0.14533
Total	0.00391	220.12128	0.00110	0.00281	0.02210	0.61788	19887.17619	556091.75972	0.00539	0.15072
Spreadsheet	0.00391	220.12128	0.00110	0.00281	0.02210	0.61788	19887.17619	556091.75972	0.00539	0.15072

The unit block Displacement and stress





The comparison of the Elmer solution with the UMAT

	Total Strain	Stress (MPa)	Elastic Strain	Plastic Strain	Plastic Work Inc (MPa)	Cum Work (MPa)	Heat Inc (J/m3)	Cum Heat (J/m3)	Temp Rise Inc (C)	Cum Temp
Spreadsheet	0.00391	220.12128	0.00110	0.00281	0.02210	0.61788	19887.17619	556091.75972	0.00539	0.15072
Elmer	0.00391	220.12000	0.00110	0.00281	0.02210	0.61788	19887.00000	556090.00000	0.00539	0.15072

The material input for the UMAT is as follows

Material 1

Number of Material Constants = Integer 8

Number of State Variables = Integer 0

! List material constants as {Young's modulus, Poisson's ratio, Hardening Slope, Yield, density, Cp, beta, npt, Tsteps, Eout}:

Material Constants(8) = Real 200000.0 0.30 0.1 220.0 7850 470 0.90 2

UMAT Subroutine = File "PDEUmat" "plastic"

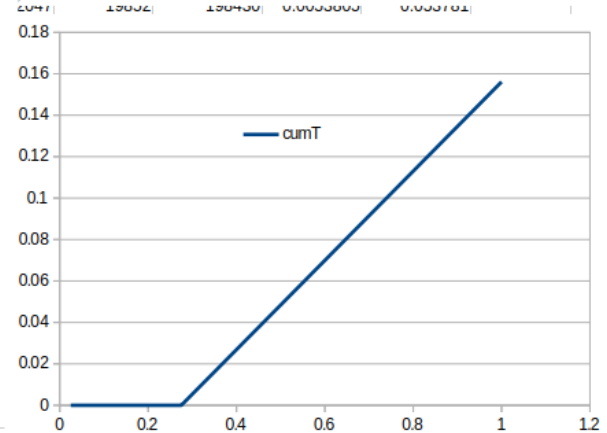
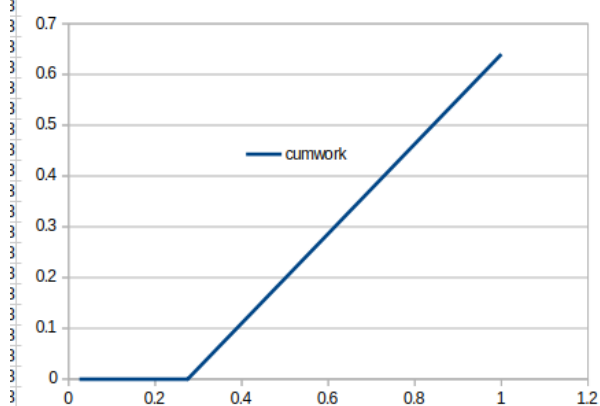
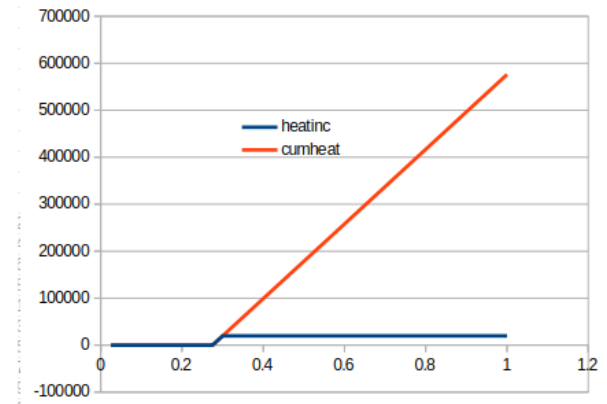
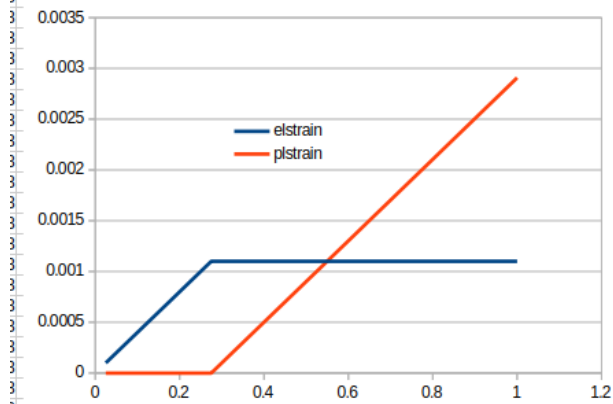
Name = "plastic" ! This specifies the CMNAME argument of UMAT

End

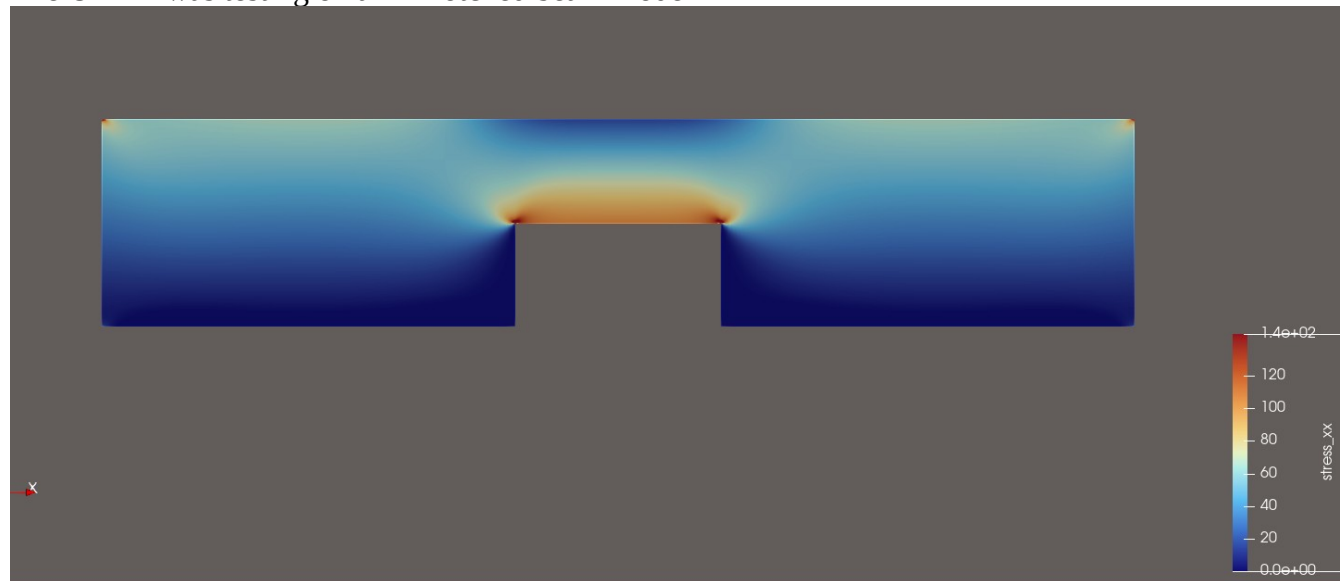
The last parameter Eout limits the printed data to 1 selected element, the point of interest where yielding is occurring. This is done to limit the size of the file. The data is calculated for all elements, but unless it is a very small model, printed output would be too large.

A future upgrade could be to add VTU output.

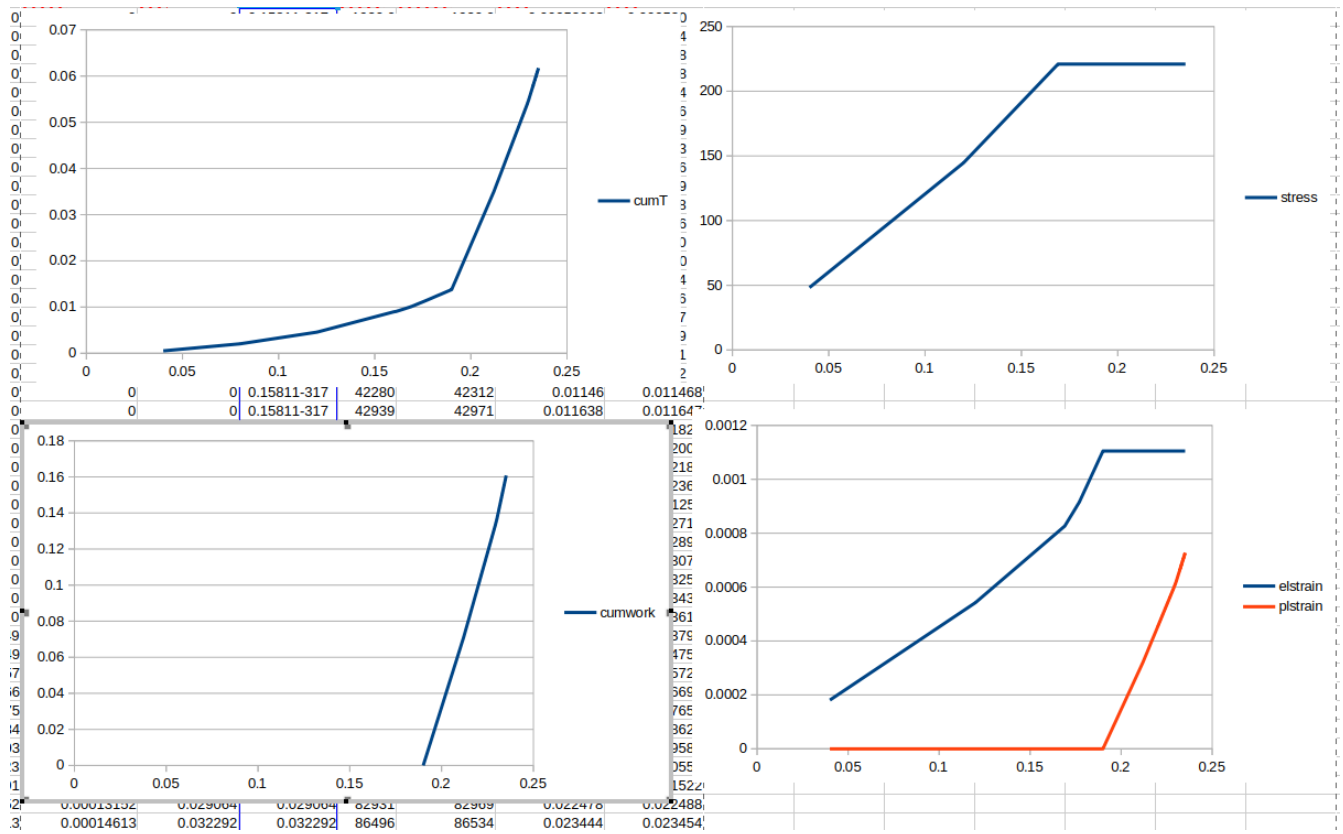
Plots of time histories from output file



The UMAT was testing on a 2D notched beam model



Notch Beam Graphs



UMAT and test case can be found at

<https://github.com/mrkearden/PlasticDisipationEnergy-Elmer-Solver>